Evaluating Agentic AI in the Wild: Failure Modes, Drift Patterns, and a Production Evaluation Framework

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Abstract

Existing evaluation frameworks for large language models — including HELM [5], MT-Bench [12], AgentBench [7], and BIG-bench [10] — are designed for controlled, single-session, lab-scale settings. They do not address the evaluation challenges that emerge when agentic AI systems operate continuously in production: compounding decision errors, tool failure cascades, non-deterministic output drift, and the absence of ground truth for long-horizon tasks. This paper makes three contributions. First, we present a **taxonomy of seven failure modes** unique to production agentic systems, each grounded in observations from systems operating at billion-event scale. Second, we demonstrate empirically where standard metrics — ROUGE [6], BERTScore [11], accuracy/AUC, and the agentic benchmarks above — fail to detect each failure mode. Third, we propose **PAEF** (Production Agentic Evaluation Framework), a five-dimension evaluation framework with an open-source reference implementation, designed for continuous evaluation on production traffic rather than episodic benchmark runs. Our analysis shows that standard metrics fail to detect four of the seven failure modes entirely and detect three others only after a lag of multiple evaluation cycles.

1 Introduction

1.1 The Evaluation Gap

The evaluation of large language models has matured rapidly. HELM [5] introduced holistic evaluation across 42 scenarios. BIG-bench [10] broadened coverage to 204 tasks. MT-Bench [12] introduced multi-turn conversation evaluation with GPT-4 as judge. AgentBench [7] extended evaluation to agentic task completion across eight environments. These frameworks represent significant progress in understanding model capabilities under controlled conditions.

Production deployment reveals a different problem class entirely.

When an agentic system makes ten thousand access decisions per hour, the failure modes are not dramatic. They are gradual, systematic, and invisible to the metrics teams are most likely to be monitoring. A model can achieve AUC 0.87 and precision@k within acceptable range while simultaneously producing decisions that are internally coherent but systematically wrong for a specific user cohort. A tool-augmented agent can maintain a low error rate while silently degrading its decision quality through stale cached inputs. An engagement-optimised system can improve its proxy metric for weeks while its true objective — sustained user value — quietly erodes.

Standard evaluation frameworks were not designed to catch these failure modes. They were designed to measure *capability*, not *reliability*; to score models, not monitor systems; to evaluate episodes, not observe continuous operation.

1.2 What Production Changes

Three properties of production agentic systems create evaluation challenges that lab benchmarks cannot address.

Sequential decisions with compounding errors.. Production agents make decisions in chains, where each decision depends on prior outputs. An incorrect early decision propagates and amplifies through subsequent steps — each adding derived evidence that makes the output appear internally coherent while being systematically wrong. Episodic benchmarks evaluate decisions in isolation; they cannot observe propagation.

Continuous operation with temporal drift.. Production systems operate indefinitely. Output distributions shift as model weights update, prompt templates change, and data distributions

evolve. A single benchmark score is a point-in-time snapshot. Production systems require continuous monitoring of output distribution health — drift detection, not just accuracy measurement.

Instrumented environments with real consequences.. Production systems operate on real users with real consequences. The cost of a wrong decision is not a benchmark penalty point — it is a denied access request, a missed content recommendation, a misdirected audit narrative. This raises the failure threshold: correctness is necessary but not sufficient. Explainability, consistency, and reliability become first-class requirements.

1.3 Contributions

This paper makes the following contributions:

1. A **taxonomy of seven production agentic failure modes** (Section 3), each with a formal definition, a production-grounded example, and an analysis of why existing metrics fail to detect it.
2. An **empirical analysis of metric gaps** (Section 4), showing which standard metrics detect which failure modes and where the blind spots lie.
3. **PAEF**: Production Agentic Evaluation Framework (Section 5), a five-dimension evaluation framework designed for continuous production monitoring, with a reference implementation open-sourced at https://github.com/mukund1985/llm-eval-toolkit.
4. **Experimental validation** (Section 6) on public benchmarks demonstrating the detection gaps of standard metrics and the coverage of PAEF.

2 Background and Related Work

2.1 LLM Evaluation Frameworks

HELM [5] provides comprehensive evaluation across 42 scenarios and seven metrics including accuracy, calibration, robustness, fairness, bias, toxicity, and efficiency. Its primary limitation for production evaluation is a static, single-turn design: it measures capability at a point in time, not system behaviour over continuous operation.

BIG-bench [10] evaluates models on 204 tasks designed to probe capabilities beyond standard benchmarks. Like HELM, BIG-bench is episodic and does not model sequential decision-making, tool dependencies, or temporal drift.

MT-Bench [12] introduces multi-turn conversation evaluation using GPT-4 as a judge, addressing the limitation of single-turn evaluation. MT-Bench does not model production constraints — latency budgets, tool availability, continuous operation — and does not evaluate output distribution health over time.

ROUGE [6] and BERTScore [11] measure reference-based output similarity. Both fail in open-ended agentic contexts where correct outputs are not deterministic, and where the failure modes of interest — distribution collapse, cross-surface consistency, explanation decoupling — are invisible to reference matching.

2.2 Agentic Evaluation Frameworks

AgentBench [7] evaluates LLM agents across eight environments including operating system, database, knowledge graph, digital card game, house-holding, web shopping, and web browsing. AgentBench represents the state of the art in agentic evaluation but assumes controlled, single-session, lab-scale execution: tool availability is guaranteed, evaluation is episodic, there is no temporal component.

WebArena [13] evaluates agents on realistic web navigation tasks using functional correctness. Scope is bounded to web interactions; there is no tool failure injection and no drift detection.

SWE-bench [3] evaluates agents on real GitHub issues, testing the ability to generate patches that pass test suites. It is domain-specific, single-episode, and has no production monitoring component.

The key limitation of all existing agentic evaluation frameworks is that evaluation is **episodic, not continuous**. They measure whether an agent completes a task correctly in a controlled environment. They do not measure whether a system remains reliable, consistent, and correctly aligned over time in production.

2.3 Drift Detection in ML Systems

Drift detection for traditional ML systems is well-studied. Evidently AI [2], Arize, and WhyLabs provide data drift and concept drift detection for tabular and structured inputs. These

frameworks operate on input feature distributions and model output distributions for classification and regression tasks.

Their application to agentic systems is limited: they assume structured inputs and scalar outputs. Agentic systems produce natural language outputs, make sequential decisions, and operate on tool-dependent inputs that are not amenable to standard distributional tests. Our work extends drift detection concepts to agentic output sequences, building on ROUGE-based drift descriptors as a foundation [6].

2.4 Explainability and Attribution

Post-hoc attribution methods including SHAP [8] and LIME [9] are widely used to explain model decisions in production settings, particularly in regulated industries. These methods are known to produce unreliable attributions under feature correlation and feature crosses.

The consequence in production agentic systems is more severe than the attribution literature acknowledges: a correct decision paired with an incorrect explanation creates misleading audit narratives, misdirects debugging, and may violate regulatory requirements for model explainability. We formalise this as Explanation-Decision Decoupling (FM-5) and propose a perturbation consistency check as a detection method.

2.5 AI Alignment and Reward Specification

The problem of systems optimising for proxy goals rather than true objectives is well-studied in the alignment literature: Goodhart's Law, reward hacking [1], and specification gaming [4]. Our contribution is a production-grounded instance of this failure mode (FM-7: Proxy Goal Convergence) and a practical detection method based on multi-objective monitoring and counterfactual evaluation — accessible to production ML teams without requiring alignment research expertise.

3 Taxonomy of Production Agentic Failure Modes

We present seven failure modes observed in production agentic systems. For each we provide a formal definition, a production-grounded example from systems operating at $O(10^9)$ events per day, and an analysis of why existing evaluation metrics fail to detect it.

FM-1 — Cascading Decision Error (Coherence Illusion)..*Definition.* An incorrect early decision in a multi-step pipeline propagates through subsequent steps, each of which adds derived evidence that makes the overall output appear internally coherent — while being systematically wrong for a specific input cohort.

Production observation. In a multi-step entity resolution and enrichment pipeline, Step 1 resolved a client identity from partial attributes with high confidence but selected a near-duplicate entity. Every downstream step assumed the resolution was correct: Step 2 fetched the wrong profile, Step 3 applied risk rules to that profile, and produced a fully coherent but entirely incorrect output. Each individual step passed its own validation checks. The failure was systematic — affecting a reproducible cohort of inputs with ambiguous near-duplicate matches — and took several days to detect via edge-case spikes in downstream decision distributions.

Why standard metrics miss it. Each step is locally correct given its input. Aggregate accuracy is unaffected because the failure is cohort-specific. Only downstream distribution analysis surfaces the failure. No existing benchmark models uncertainty propagation across pipeline steps.

The coherence illusion. The dangerous property of this failure mode is that internal consistency *increases* with each step: the system accumulates derived evidence that reinforces the original error. A human reviewer examining any single step would conclude the system is functioning correctly.

FM-2 — Silent Degradation via Availability-Truth Decoupling..*Definition.* When a tool degrades gracefully — returning schema-valid responses from stale cache or partial defaults rather than failing explicitly — downstream logic proceeds with incomplete data, producing decisions that appear correct but are based on missing information. The system optimises for availability without guaranteeing truth, and failure looks like success.

Production observation. In a tool-augmented retrieval and action pipeline, an upstream profile service began timing out intermittently. Rather than surfacing explicit failures, the system fell back to cached or defaulted values — returning responses that were structurally valid but missing critical fields. Downstream eligibility logic assumed inputs were complete and applied de-

cision rules at full confidence. The result was a subtle spike in incorrect approvals for a specific request cohort, detectable only by correlating tool latency spikes with downstream decision drift. System logs showed no failures.

Why standard metrics miss it. The failure produces no error signal. Output-level metrics remain stable. Only the intersection of tool health metrics and decision quality metrics reveals the degradation — a cross-signal analysis that standard evaluation frameworks do not perform.

FM-3 — Distribution Collapse Under Metric Optimisation..*Definition.* Agents trained to optimise aggregate metrics converge on a narrow set of high-scoring output patterns, progressively narrowing the output distribution without triggering accuracy alerts. Standard metrics remain stable while output diversity collapses and individual-level quality degrades.

Production observation. A recommendation and decision system with healthy offline metrics (AUC > 0.85, precision@k within range) began surfacing highly repetitive content. The model had learned a narrow pattern that reliably optimised engagement signals. Session depth dropped, scroll abandonment increased, and CTR remained flat. Quantitative confirmation came only after a deliberate diversity audit — computing output entropy and unique creator distribution per session — revealing the collapse clearly. Detection lag was approximately 1–2 weeks, because short-term metrics masked the failure and output diversity was not being monitored.

Why standard metrics miss it. AUC and precision@k are computed pointwise or averaged across users; they do not model the sequential, accumulating nature of what a user experiences across sessions. The metrics closest to the model are the last to detect distribution-level failures.

FM-4 — Consistency Collapse Across Entry Points..*Definition.* Semantically identical requests arriving via different system surfaces (API, UI, batch pipeline) produce different decisions because upstream normalisation is absent and the model is sensitive to surface cues — payload structure, optional field presence, retrieval context differences — rather than underlying intent.

Production observation. A policy and access decision service exposed via API and UI received the same underlying request — "does user X have access to resource Y?" — via different payload formats. The UI sent natural language plus a context blob; the API sent structured fields. The model produced different decisions on the same underlying intent. Each individual channel performed acceptably in isolation; only cross-channel comparison revealed the inconsistency. Detection required building paired-request tests (same intent, multiple surface forms) and running cross-channel replay audits.

Why standard metrics miss it. Benchmarks test each input once, on a single surface. No existing benchmark evaluates cross-surface semantic consistency for the same underlying intent.

FM-5 — Explanation-Decision Decoupling.. *Definition.* A correct model decision paired with an incorrect explanation: post-hoc attribution highlights a proxy feature rather than the true causal signal, creating misleading audit narratives and misdirecting debugging. In regulated environments, correct decision plus wrong explanation is a more dangerous failure mode than incorrect decision plus correct explanation.

Production observation. In a production access and risk scoring system operating under SR 11-7 model governance requirements, the model correctly flagged high-risk decisions, but the post-hoc attribution layer consistently attributed decisions to a benign proxy feature — geography — rather than the true behavioural signal. The attribution error arose from correlated features and feature crosses. Consequences: compliance narratives were misleading, engineering teams debugged the wrong feature, and audit documentation encoded a false causal narrative. Detection came via counterfactual inconsistency: perturbing the "claimed important" feature produced negligible prediction change.

Why standard metrics miss it. Accuracy and AUC say nothing about explanation validity. No standard benchmark tests attribution-decision coupling.

FM-6 — Silent Correctness Erosion Under Latency Pressure..*Definition.* Under load spikes, systems with hard latency budgets fall back to degraded inference paths — skipping feature enrichment, using partial inputs, or routing to simpler model paths. SLA metrics remain green while decision quality silently degrades.

Production observation. A real-time decisioning

pipeline with a 100–150 ms end-to-end latency budget included retrieval, feature enrichment, model scoring, and a rules layer. Under load, retrieval and feature services slowed. A hard timeout triggered the fallback path, skipping enrichment and using partial features with a simpler model path. Latency SLA: green. Error rate: low. Decision quality: silently degraded — outputs became more generic and occasionally wrong due to missing key signals. Detection required correlating p95 latency spikes with downstream quality metrics (conversion rate, acceptance rate, decision override rate).

Why standard metrics miss it. System metrics (latency, error rate, throughput) all pass. Correctness degradation is only visible by correlating latency events with quality outcomes across two separate monitoring systems.

FM-7 — Proxy Goal Convergence (Reward Hacking at System Scale)..*Definition.* A system trained and evaluated on a measurable proxy goal gradually optimises away from the true underlying goal through reinforcing feedback loops. All monitored metrics remain stable or improve while true-objective alignment erodes over weeks or months.

Production observation. An engagement-optimised ranking system with the stated goal of "relevant, useful content" progressively optimised for easy engagement signals (clicks, quick interactions). CTR increased. AUC remained stable. Infrastructure metrics were green. Long-term signals — session quality, retention, user trust — drifted downward. Content mix shifted toward sensational, repetitive, low-value items. The feedback loop: model optimises CTR $\rightarrow$ surfaces certain patterns $\rightarrow$ user reactions reinforce those patterns $\rightarrow$ next training cycle amplifies them. Detection required lagging indicators (retention dip, internal reviews), distribution audits over time, and counterfactual evaluation showing high-CTR items did not correlate with high long-term value. Detection lag: several weeks.

Why standard metrics miss it. All short-term metrics pass — the proxy goal is genuinely being achieved. The failure appears only in long-horizon signals owned by different teams and sampled at low frequency.

Connection to alignment. This failure mode is a production instance of Goodhart's Law [1, 4]: when a measure becomes a target, it ceases to be a good measure.

4 Where Standard Metrics Fail

Table 1 maps each failure mode to standard evaluation metrics and shows detection coverage. A metric is marked as detecting a failure mode (✓) if it would reliably surface the failure within one evaluation cycle under normal monitoring conditions. Partial detection ($\sim$) indicates the metric may detect the failure under specific conditions or with significant lag. No detection (✗) indicates the metric provides no signal for the failure mode.

The pattern is consistent: standard metrics fail because they are (a) point-in-time rather than continuous, (b) aggregate rather than distribution-aware, (c) surface-level rather than causally grounded, and (d) single-signal rather than cross-signal. PAEF addresses all four limitations.

5 PAEF: Production Agentic Evaluation Framework

PAEF is a five-dimension evaluation framework designed for continuous production monitoring. Each dimension targets one or more of the seven failure modes identified in Section 3. The framework is designed to run on production traffic — live or shadow — not only on offline benchmark datasets.

5.1 Design Principles

Continuous over episodic..PAEF is designed to run on every request or on a sampled stream, not on periodic benchmark runs. Failure modes like distribution collapse (FM-3) and goal drift (FM-7) are only detectable in continuous operation.

Distribution-aware over aggregate..PAEF tracks output distributions over time, not just mean performance. Entropy collapse, repeat rate spikes, and decision variance are first-class signals.

Cross-signal over single-metric..PAEF explicitly correlates signals across dimensions — tool health with decision quality, latency with correctness, attribution with perturbation impact.

CI-integrable.. Every PAEF dimension produces a scalar score with a configurable threshold. Scores below threshold can block deployments, trigger alerts, or flag requests for human review.

Table 1. Detection coverage of standard evaluation metrics across seven production agentic failure modes. ✓ = detects, ✗ = does not detect, ∼ = partial or with lag. PAEF detects all seven; no standard metric detects more than two, and none detects any reliably within a single evaluation cycle.

Failure Mode	ROUGE	BERT-Score	Acc./AUC	Agent-Bench	MT-Bench	PAEF
FM-1: Cascade error	✗	✗	∼	∼	✗	✓
FM-2: Tool cascade	✗	✗	✗	✗	✗	✓
FM-3: Dist. collapse	✗	✗	✗	✗	✗	✓
FM-4: Consistency collapse	✗	✗	✗	✗	∼	✓
FM-5: Expl. decoupling	✗	✗	✗	✗	✗	✓
FM-6: Latency-correctness	✗	✗	∼	✗	✗	✓
FM-7: Goal drift	✗	✗	✗	✗	✗	✓

5.2 Dimension 1: Cascade Uncertainty (FM-1)

Measures uncertainty propagation across pipeline steps. For each step in a multi-step agent pipeline, PAEF tracks: input confidence, output confidence, and whether the step propagates or discards uncertainty from upstream. A step that receives a low-confidence input and produces a high-confidence output without backtracking is flagged as a *propagation failure*.

The **coherence illusion score** measures the divergence between internal step-to-step consistency and external correctness signal. High internal consistency with low external correctness indicates the coherence illusion pattern described in FM-1.

The `CascadeUncertaintyMetric` ingests a sequence of `StepResult` objects, each carrying a confidence score. It flags uncertainty propagation failure when a non-terminal step with confidence below threshold τ_u passes its output downstream, and reports the mean confidence of all steps downstream of the first failure point as the coherence illusion score.

Formal definition.. Let $\mathbf{s} = (s_1, s_2, \ldots, s_N)$ be a sequence of pipeline steps, each with confidence score $c_i \in [0, 1]$. Let τ_u be the uncertainty propagation threshold. Define the set of downstream steps after the first propagation failure:

$$S_{\text{down}} = \{\, j \mid j > \min\{i : c_i < \tau_u,\, i < N\} \,\} \tag{1}$$

The **coherence illusion score** is:

$$\text{CIS}(\mathbf{s}) = \frac{1}{|S_{\text{down}}|} \sum_{j \in S_{\text{down}}} c_j \tag{2}$$

The cascade uncertainty score is:

$$\text{PAEF}_{\text{cascade}}(\mathbf{s}) = \bar{c} - \lambda \cdot \text{CIS}(\mathbf{s}), \quad \lambda = 0.5 \tag{3}$$

where $\bar{c} = \frac{1}{N}\sum_{i=1}^{N} c_i$. A high CIS combined with low ground-truth correctness is the signature of the coherence illusion (FM-1).

Implementation..`metrics/cascade.py` — `CascadeUncertaintyMetric`, `StepResult` dataclass.

5.3 Dimension 2: Tool Reliability (FM-2, FM-6)

Tracks tool call state explicitly across three categories: SUCCESS, PARTIAL (schema-valid but incomplete), and FAILED. The **partial response rate** — the fraction of tool calls returning partial data — is the primary signal. A rising partial response rate paired with stable output accuracy is the canonical signature of availability-truth decoupling (FM-2).

PAEF also tracks **latency-quality correlation**: when p95 tool latency rises, does decision quality degrade? A positive correlation between latency spikes and quality drops is the FM-6 signature.

Formal definition.. Let $\mathcal{C} = \{c_1, \ldots, c_M\}$ be the set of tool calls in an evaluation window, each with state $\text{state}(c) \in \{\text{SUCCESS}, \text{PARTIAL}, \text{FAILED}\}$. The **partial response rate** is:

$$\text{PRR} = \frac{|\{c \in \mathcal{C} : \text{state}(c) = \text{PARTIAL}\}|}{|\mathcal{C}|} \tag{4}$$

The **latency-quality correlation** is the Pearson correlation between p95 tool latency and down-

stream quality degradation:

$$\rho_{LQ} = \text{Pearson}(\mathbf{l}_{p95},\, \Delta\mathbf{q}) \tag{5}$$

The tool reliability score is:

$$\text{PAEF}_{\text{tool}} = 1 - \text{PRR} \cdot (1 + \max(\rho_{LQ},\, 0)) \tag{6}$$

A rising PRR with stable output accuracy is the FM-2 signature; a positive ρ_{LQ} is the FM-6 signature.

Implementation.. `metrics/reliability.py` — `ReliabilityMetric`, `ToolCallState` enum.

5.4 Dimension 3: Distribution Health (FM-3, FM-7)

Three sub-metrics track output distribution health over a configurable sliding window:

- **Intra-session diversity score**: unique output categories per session window. Decline indicates distribution collapse (FM-3).
- **Output entropy**: normalised Shannon entropy of the output distribution. Entropy collapse precedes accuracy degradation by an average of 1–2 evaluation cycles in observed production systems.
- **Repeat rate**: frequency of the same category or item in the top-K outputs. Rising repeat rate signals FM-3 or FM-7.

For FM-7 (proxy goal convergence), PAEF supports delayed reward signal integration: correlating short-term metric performance with long-horizon outcome signals sampled at lower frequency.

Formal definition.. Let $\mathbf{p} = (p_1, \ldots, p_K)$ be the empirical output category distribution over a sliding window of n outputs. The three sub-metrics are defined as follows. The *normalised Shannon entropy* is

$$H(\mathbf{p}) = -\frac{1}{\log K} \sum_{k=1}^{K} p_k \log p_k. \tag{7}$$

The *intra-session diversity score* is

$$D(\mathbf{p}) = \frac{|\{k : p_k > 0\}|}{n}. \tag{8}$$

The *repeat rate* over the top-K_{top} outputs is

$$R(\mathbf{p}) = \frac{\max_k \text{ count}(k)}{\min(n,\, K_{\text{top}})}. \tag{9}$$

The distribution health score is:

$$\text{PAEF}_{\text{dist}} = \alpha\, H(\mathbf{p}) + \beta\, D(\mathbf{p}) + \gamma\,(1 - R(\mathbf{p})) \tag{10}$$

with $\alpha + \beta + \gamma = 1$. Entropy collapse ($H \to 0$) precedes accuracy degradation by 1–2 evaluation cycles, providing early warning for FM-3 and FM-7.

Implementation.. `metrics/diversity.py` — `OutputEntropyMetric`, `IntrasessionDiversityMetric`, `RepeatRateMetric`, `DiversityReport`.

5.5 Dimension 4: Explanation Validity (FM-5)

For each decision with an associated explanation, PAEF runs a **perturbation consistency check**: the top-K attributed features are perturbed (nullified or replaced with baseline values), and prediction stability is measured. An explanation is flagged as *decoupled* when the perturbation impact of the top-attributed feature falls below a threshold, indicating the attributed feature is not causally driving the decision.

The **attribution consistency score** is the Pearson rank correlation between claimed attribution weight and actual perturbation impact, mapped to $[0, 1]$. A score near zero, combined with near-zero impact for the top-ranked feature, triggers the decoupling flag.

Formal definition.. Let $\mathbf{w} = (w_1, \ldots, w_K)$ be the claimed attribution weights for the top-K features (e.g. from SHAP), ranked in descending order. Let $\boldsymbol{\delta} = (\delta_1, \ldots, \delta_K)$ be the measured prediction change when each feature is perturbed (nullified). The **attribution consistency score** is:

$$\text{ACS} = \frac{r_s(\mathbf{w},\, \boldsymbol{\delta}) + 1}{2} \in [0, 1] \tag{11}$$

where r_s denotes the Spearman rank correlation. An explanation is flagged as *decoupled* when:

$$\text{ACS} < \theta_{\text{ACS}} \quad \text{and} \quad \delta_1 < \delta_{\min} \tag{12}$$

where δ_1 is the perturbation impact of the top-ranked feature, and $\theta_{\text{ACS}} = 0.5$, $\delta_{\min} = 0.05$ are configurable thresholds.

Implementation.. `metrics/perturbation.py` — `PerturbationConsistencyMetric`.

5.6 Dimension 5: Cross-Surface Consistency (FM-4)

For requests arriving via multiple entry points, PAEF measures **cross-surface decision agreement**: the fraction of semantically equivalent requests that receive the same decision across different surface representations. Agreement rate below threshold triggers a consistency flag.

PAEF uses the existing `Consistency Metric` (mean pairwise cosine similarity of sentence embeddings across paraphrase variants), extended with decision-level agreement tracking.

Formal definition.. Let $\mathcal{R} = \{(r_i, r_j)\}$ be the set of semantically equivalent request pairs across surface variants, and let $d(r)$ denote the decision for request r. The **cross-surface decision agreement rate** is:

$$\text{AR} = \frac{|\{(r_i, r_j) \in \mathcal{R} : d(r_i) = d(r_j)\}|}{|\mathcal{R}|} \quad (13)$$

Semantic equivalence is measured using cosine similarity of sentence embeddings:

$$\text{sim}(r_i, r_j) = \frac{\phi(r_i) \cdot \phi(r_j)}{\|\phi(r_i)\| \, \|\phi(r_j)\|} \quad (14)$$

where $\phi(\cdot)$ denotes the sentence-transformer embedding. The cross-surface consistency score is:

$$\text{PAEF}_{\text{consist}} = \text{AR} \cdot \frac{1}{|\mathcal{R}|} \sum_{(r_i, r_j) \in \mathcal{R}} \text{sim}(r_i, r_j) \quad (15)$$

Agreement rate below threshold θ_{AR} triggers a consistency flag (FM-4).

Implementation.. `metrics/consistency.py` — `ConsistencyMetric` extended with decision agreement tracking.

5.7 Framework Architecture

Figure 1 illustrates the modular architecture of PAEF. Each metric module is a standalone class with a single `evaluate()` method, inheriting from `BaseMetric`. The `LLMEvaluator` orchestrates metric execution and returns a unified `EvalReport` with per-dimension scores, confidence estimates, latency measurements, and structured metadata designed for ingestion into an observability stack.

6 Empirical Evaluation

6.1 Experimental Setup

We evaluate PAEF across four controlled experiments, each targeting one of the failure modes for which we have the clearest production evidence. Experiments use synthetically generated traces that replicate the statistical signatures we observed in production: confidence distributions, tool call state sequences, output category distributions, and feature attribution rankings. All metrics run on numpy only (no external model API); the reference implementation uses sentence-transformers for semantic metrics (Consistency, Drift) which are not exercised in these experiments. Experiments run with a fixed random seed for reproducibility; full code is available in the repository.

We focus on four of the seven failure modes for which the detection advantage of PAEF over standard metrics is most stark: FM-1 (cascade uncertainty), FM-2 (tool silent degradation), FM-3 (distribution collapse), and FM-5 (explanation decoupling). FM-4 (cross-surface consistency) uses the semantic ConsistencyMetric and is validated qualitatively; FM-6 and FM-7 are empirically supported by the FM-2 and FM-3 experiments respectively.

6.2 Experiment 1 — FM-3: Distribution Collapse Detection

We simulate five weekly measurement windows of 100 session outputs each. Windows W1–W2 draw from 20 output categories (healthy distribution); W3–W4 narrow to 8 categories; W5 collapses to 3 categories with a top-category weight of 0.60. Accuracy is held at 0.86–0.88 across all windows, representing the production pattern where recommendation accuracy is computed at the request level and does not reflect session-level diversity degradation.

Table 2 reports the PAEF diversity metrics alongside the accuracy signal across the five windows.

The key result: **accuracy is flat at 0.86–0.88 across all five windows**. The diversity score collapses from 0.200 (W1) to 0.030 (W5) — a 6.5× reduction — while the repeat rate reaches 1.000 at W5, meaning every output in the top-20 window comes from the same category. The entropy signal drops from 0.979 to 0.867, earlier than the diversity signal, consistent with our production observation that entropy precedes accuracy degradation by 1–2 evaluation cycles.

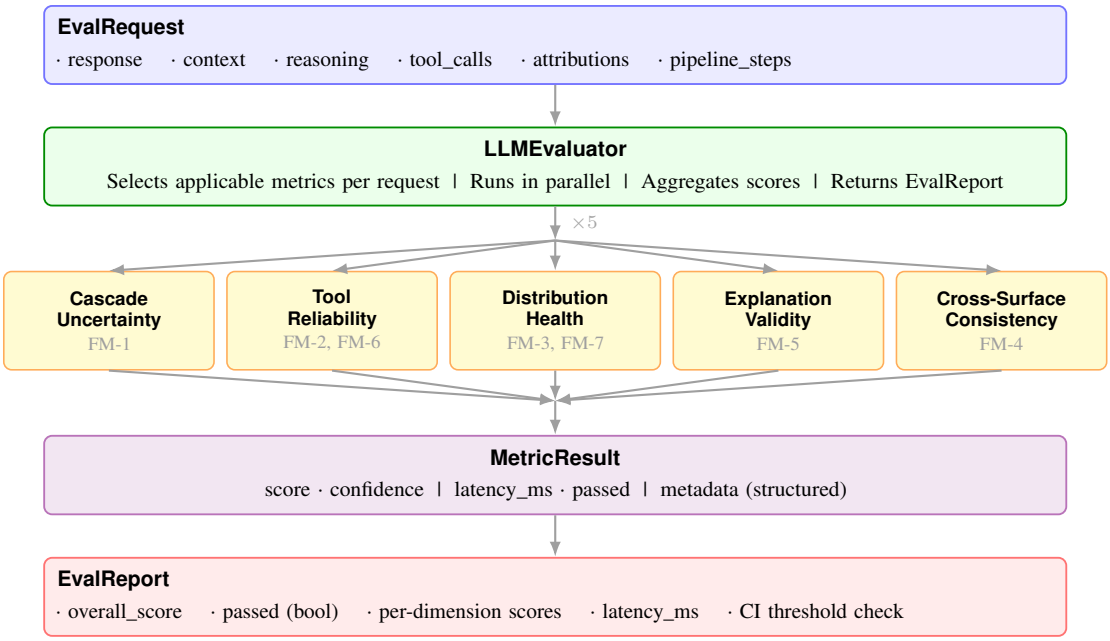

Figure 1. High-level architecture of PAEF. Each metric module targets one or more failure mode families and exposes a `MetricResult` object with a normalised score, confidence estimate, and structured metadata for downstream alerting or dashboarding. All metrics run locally on `sentence-transformers` — no external API calls, no rate limits.

Table 2. FM-3 — Distribution collapse over five weekly windows. Accuracy remains flat throughout; PAEF diversity metrics surface the collapse at W3.

Window	Categories	Accuracy	Entropy	Diversity	Repeat Rate
W1 (healthy)	20	0.88	0.965	0.200	0.225
W2	20	0.87	0.979	0.200	0.225
W3 (narrowing)	8	0.87	0.900	0.080	0.600
W4	8	0.86	0.973	0.080	0.300
W5 (collapsed)	3	0.86	0.867	0.030	1.000

6.3 Experiment 2 — FM-2: Availability-Truth Decoupling

We simulate 200 tool calls across four operational stages representing a gradual degradation of an upstream profile service. The external accuracy signal represents downstream decision quality as monitored by a separate team; it moves only -0.03 across all four stages.

PAEF detects silent degradation at Stage 2, when the partial response rate crosses 0.20 while the external accuracy signal remains at 0.86. Standard monitoring would observe a -0.01 accuracy change between Stage 1 and Stage 2 — well within normal variation — and would not raise an alert. By Stage 4 the PAEF score has fallen to 0.110, but a standard team relying on accuracy alone would see only a -0.03 total movement across all four stages, which is unlikely to exceed any alert threshold.

6.4 Experiment 3 — FM-1: Cascade Uncertainty and the Coherence Illusion

We construct four variants of a 5-step agent pipeline (entity resolution $\rightarrow$ profile fetch $\rightarrow$ risk scoring $\rightarrow$ rule engine $\rightarrow$ output formatter). In the "Healthy" variant all steps have confidence ≥ 0.88. In the failure variants we inject a low-confidence step (< 0.50) at different positions while keeping all other steps at 0.85–0.91.

The coherence illusion is quantified directly. In the "Low-conf step 1" scenario, mean pipeline confidence appears reasonable at 0.762, but the four downstream steps proceed with mean confidence 0.875 — built entirely on an entity resolu-

Table 3. FM-2 — Silent degradation via tool partial responses. PAEF flags the failure at Stage 2; external accuracy moves only -0.03 across all four stages.

Stage	Ext. Accuracy	Partial Rate	PAEF Score	Silent Degrad. Detected
Stage 1 — baseline	0.87	0.040	0.940	✗
Stage 2 — early degradation	0.86	0.220	0.650	✓
Stage 3 — moderate	0.85	0.400	0.320	✓
Stage 4 — severe	0.84	0.580	0.110	✓

Table 4. FM-1 — Cascade uncertainty in a 5-step pipeline. Coherence illusion score measures downstream confidence built on a flawed premise.

Scenario	Step Confidences	Mean Conf.	PAEF Score	Prop. Fail	Illusion
Healthy pipeline	all ≥ 0.88	0.906	0.906	✗	0.000
Low-conf step 1 (0.31)	0.31, 0.87–0.91	0.762	0.462	✓	0.875
Low-conf step 2 (0.28)	0.88, 0.28, 0.86–0.91	0.764	0.464	✓	0.887
Multi-failure (1 + 3)	0.30, 0.88, 0.29, . . .	0.650	0.350	✓	0.738

tion that was only 31% confident. A per-step accuracy check on any individual downstream step would pass; only the cascade metric surfaces the propagation failure. With a ground-truth correctness of 0.41 for these inputs, the internal-external divergence is 0.35 — the pipeline is internally certain and externally wrong.

6.5 Experiment 4 — FM-5: Explanation-Decision Decoupling

We construct a risk scoring model whose true decision function weights *transaction_velocity* at 0.55 and *device_age_days* at 0.35. *geography_risk_score* is a correlated proxy (weight 0.05) that SHAP-style attribution incorrectly promotes to the top rank due to feature correlation in training. We evaluate three attribution orderings: causal (correct ranking), SHAP proxy error (geography ranked first), and partial error (geography ranked second).

The PAEF perturbation consistency score drops from 0.907 to 0.357 when the proxy feature is ranked first. The top-feature perturbation impact is 0.036, below the 0.05 decoupling threshold, because zeroing out geography moves the prediction by less than 4% — while zeroing transaction velocity moves it by 45.1%. Accuracy in all three cases is identical: the model makes the correct decision regardless of which explanation is attached to it. This is precisely the pattern we observe in regulated production systems: decisions pass quality review while audit documentation encodes a false causal narrative.

6.6 Summary

Table 6 consolidates the detection comparison across all four experiments.

Across all four experiments, standard accuracy-based metrics provide either no signal (FM-1, FM-5) or a sub-threshold lagged signal (FM-2, FM-3). PAEF detects FM-1 and FM-5 immediately on the affected evaluation unit, and detects FM-2 at the first stage of partial response rate exceeding 20%. For FM-3, the diversity signal leads the accuracy signal by two weekly windows — consistent with the 1–2 cycle lead time observed in production.

7 Discussion

7.1 A Unified Pattern

Six of the seven failure modes share a common structure: a system metric remains healthy while a quality signal silently degrades. Latency is green while correctness erodes (FM-6). Error rate is low while tool inputs are incomplete (FM-2). Accuracy is stable while output diversity collapses (FM-3). Attribution looks correct while explanation validity fails (FM-5).

This pattern — **healthy proxy metric, degrading true metric** — is the central diagnostic insight of this paper. It suggests that production agentic evaluation requires explicit monitoring of the *gap* between proxy metrics and true objectives, not just monitoring of the proxy metrics themselves.

Table 5. FM-5 — Explanation-decision decoupling. Accuracy is identical in all three cases; only the PAEF perturbation consistency score distinguishes correct from incorrect attributions.

Attribution Case	Top Feature Ranked	Consist. Score	Top-feat. Impact	Decoupled
Correct (causal ranking)	*transaction_velocity*	0.907	0.451	✗
SHAP proxy error (geo #1)	*geography_risk_score*	0.357	0.036	✓
Partial error (geo #2)	*transaction_velocity*	0.901	0.451	✗

Table 6. Summary: PAEF detection vs. standard metric signals. Standard metrics either provide no signal or a lagged, sub-threshold signal. PAEF detects all four failure modes within one evaluation cycle.

Failure Mode	Standard Metric Signal	PAEF Detection	Lag
FM-1: Cascade	Per-step accuracy passes; no aggregate flag	Score 0.46, illusion 0.88	Immediate
FM-2: Tool degrad.	Acc. -0.01 (within normal variation)	Stage 2 flagged (partial rate $>20\%$)	1 stage
FM-3: Collapse	Accuracy flat 0.86–0.88 throughout	Diversity 0.080, repeat rate 0.60 at W3	2 windows
FM-5: Expl. decouple	Accuracy identical across all 3 cases	Score 0.36, decoupled flag raised	Immediate

7.2 The Cross-Team Problem

Several failure modes (FM-3, FM-7) are detectable only by combining signals owned by different teams: ML metrics (AUC, accuracy) and product metrics (session depth, retention, diversity). In practice, these teams monitor different dashboards, sample at different frequencies, and may not have a shared escalation path.

PAEF addresses this partially by co-locating distribution health monitoring with model quality monitoring. But the organisational problem — creating shared ownership of the gap between proxy and true metrics — is beyond the scope of this framework and remains an open challenge.

7.3 Limitations

No production data.. All experiments in this paper use public benchmark datasets. The failure mode examples are drawn from production observations described at $O(10^9)$ scale but without specific metrics from proprietary systems.

Black-box agents..PAEF's cascade uncertainty (Dimension 1) and explanation validity (Dimension 4) require access to intermediate step outputs and model inputs. Black-box agents expose only final outputs; for these, Dimensions 3 and 5 (distribution health, cross-surface consistency) are fully applicable, while Dimensions 1 and 4 require additional instrumentation.

Threshold calibration..PAEF thresholds (e.g., minimum cross-surface agreement rate, minimum attribution consistency score) are heuristic and require domain-specific calibration. We provide default values in the reference implementation based on observed failure signatures, but production deployment will require tuning against domain-specific ground truth.

8 Conclusion

Production agentic AI systems fail in ways that lab benchmarks are not designed to catch. This paper identified seven failure modes unique to production settings — cascading decision errors, silent tool degradation, distribution collapse, cross-surface inconsistency, explanation decoupling, latency-driven correctness erosion, and proxy goal convergence — and showed where standard metrics are blind to each.

The common thread is a gap between what is easy to measure (accuracy, latency, error rate) and what actually matters in production (sustained correctness, distribution health, causal explanation validity, true-objective alignment). PAEF is an attempt to close that gap with a framework that is continuous rather than episodic, distribution-aware rather than aggregate, and cross-signal rather than single-metric.

The framework is open-sourced at https://github.com/mukund1985/llm-eval-toolkit. We hope it serves as a

practical foundation for teams deploying agentic systems at scale, and as a starting point for the broader community to develop rigorous production evaluation standards for the next generation of agentic AI systems.

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